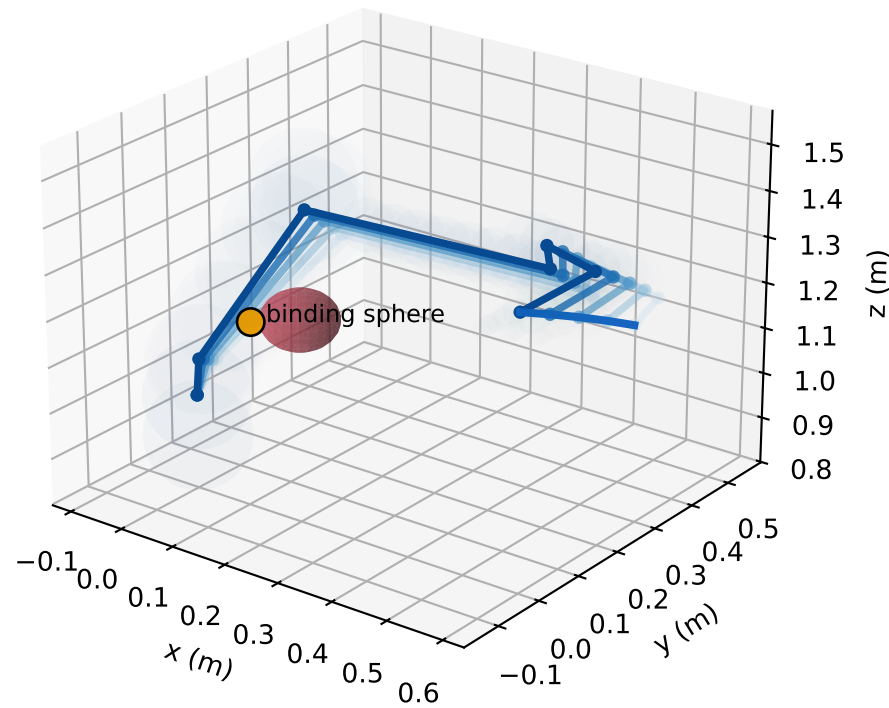
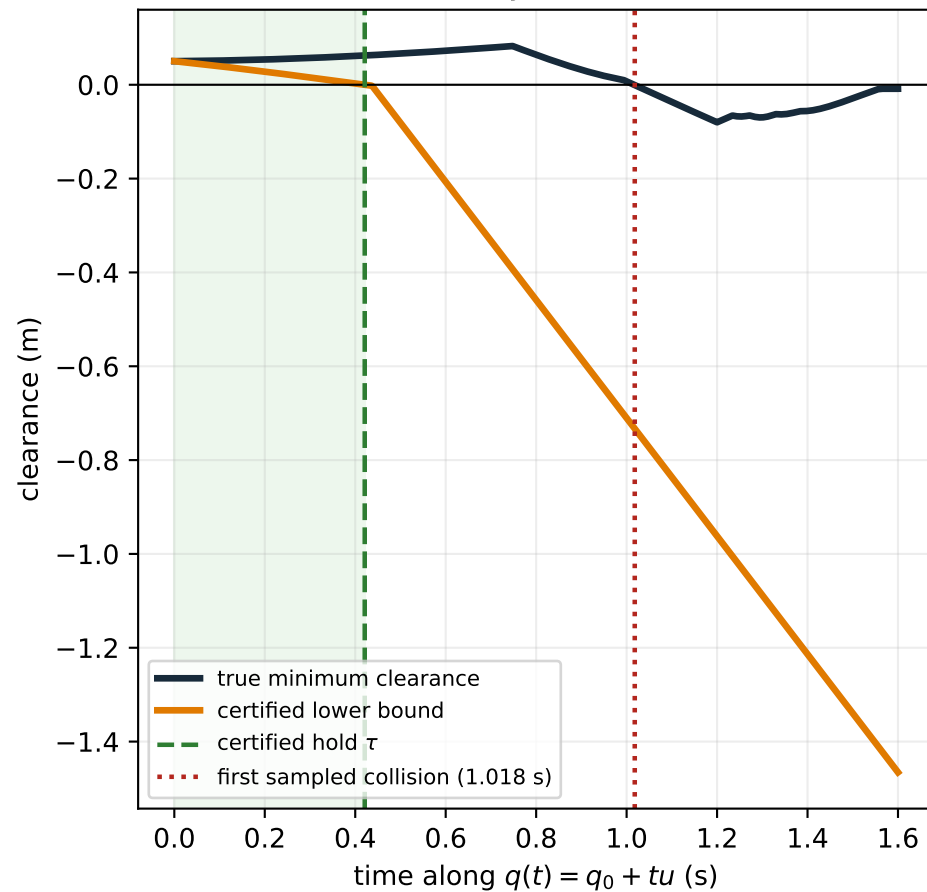


UR5 kinematic-envelope hold certificate — computed from URDF geometry

Exact UR5 motion, 0 to certified $\tau = 0.421$ s



The lower bound stays below true clearance



$D(t) \geq \|p(t) - p(0)\|$ and $D(\tau) = h_0$

